



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Hemanshu Mandalia
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Outline

1 Executive Summary

2 Introduction

3 Methodology

4 Results

5 Conclusion

Executive Summary

Summary of methodologies

- Data Collection through API
- Data Collection with Web Scraping
- Data Wrangling
- Exploratory Data Analysis using SQL
- Data Visualization using Pandas and Matplotlib
- Interactive Visual Analytics with Folium
- Interactive Dashboard with Plotly Dash
- Prediction through Machine Learning

Summary of all results

- Exploratory Data Analysis result
- Interactive analytics in screenshots
- Predictive Analytics result

Introduction

Project Background and Context

SpaceX has disrupted the aerospace industry by advertising Falcon 9 launches at \$62M, compared to \$165M+ charged by traditional providers. This 60% cost reduction is driven by first-stage reusability. By analyzing publicly available Falcon 9 launch data from SpaceX's website, we can build a predictive model to determine whether the first stage will land successfully — a key driver of launch cost. This insight enables competing launch providers like SpaceY to accurately estimate and bid against SpaceX pricing.

Core Problem Statement

Develop a machine learning model to predict if the Falcon 9 first stage will land successfully, using historical launch features such as payload mass, orbit, launch site, and booster version — enabling cost estimation and competitive bidding strategy.

SpaceX Price

\$62M

Competitors

\$165M+

Savings from Reuse

~60%

Section 1

Methodology

Methodology

01

Data Collection

API & Web Scraping: Gathered Falcon 9 launch data from SpaceX API and Wikipedia via web scraping for complete mission history

Describe how data was collected

02

Data Wrangling

Cleaned, transformed, and encoded launch data – handling missing values, one-hot encoding landing outcomes, and feature engineering

Describe how data was processed

03

EDA with SQL & Visualization

Exploratory analysis using SQL queries and Pandas/Matplotlib to identify trends in launch sites, payload, orbits, and success rates

SQL + Visual Insights

04

Interactive Visual Analytics

Built Folium maps for launch site proximity and Plotly Dash dashboard for dynamic filtering of launch outcomes

Folium & Plotly Dash

05

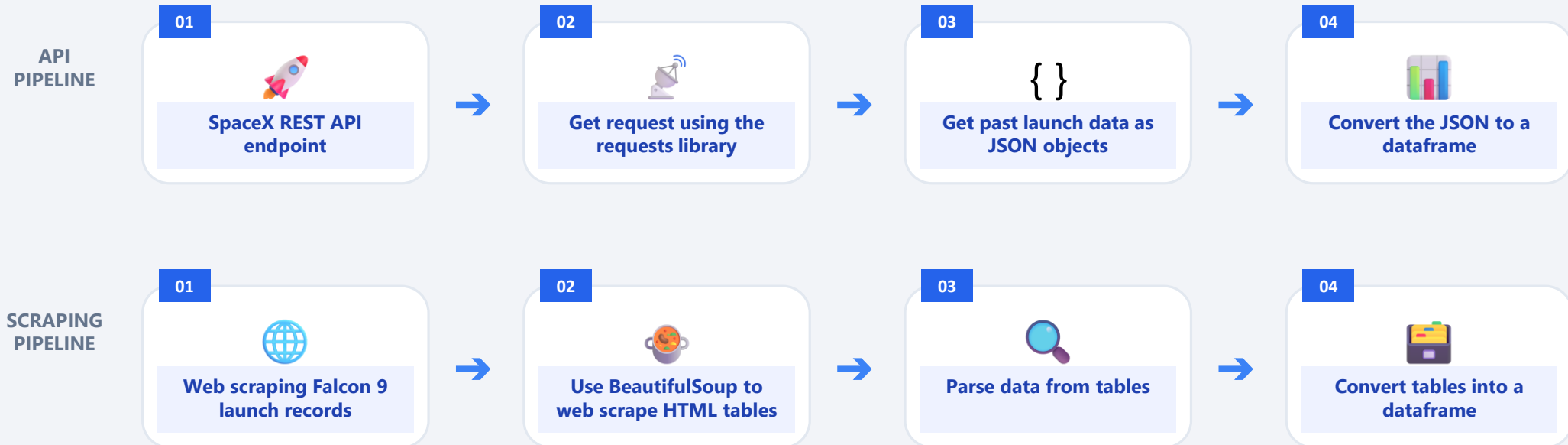
Predictive Analysis

Built, tuned & evaluated classification models (Logistic Regression, SVM, Decision Tree, KNN) to predict first-stage landing success

Build, tune, evaluate models

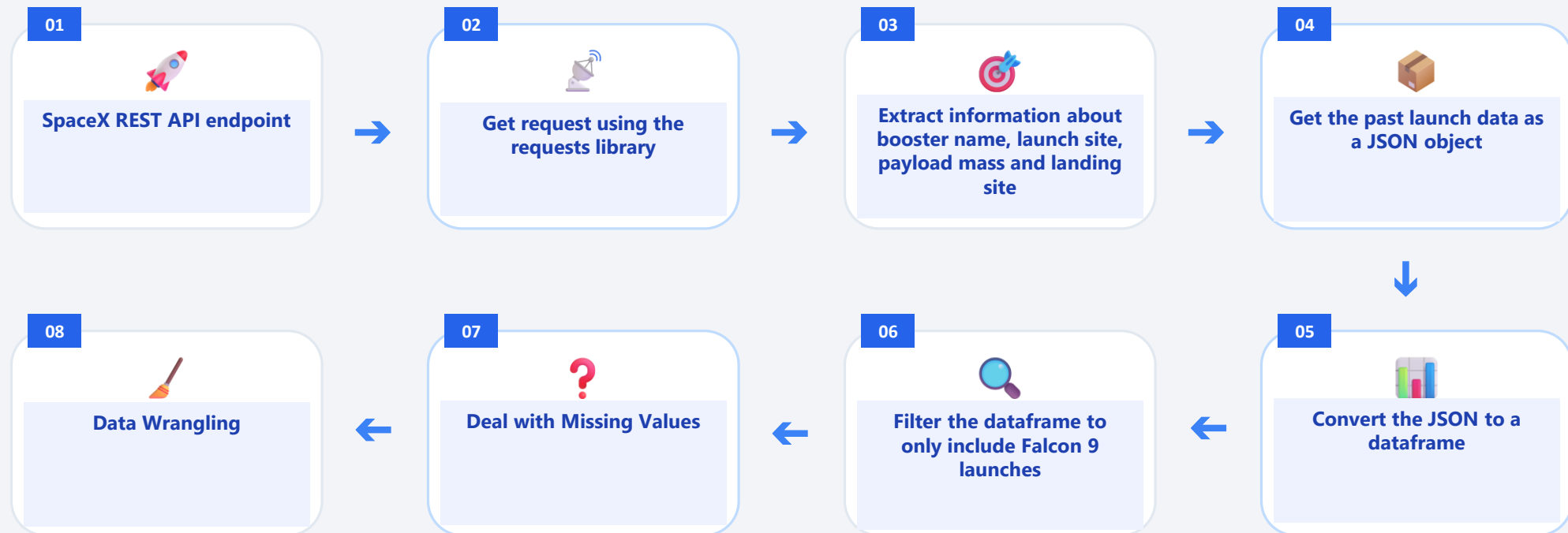
Data Collection

The data was gathered from the spaceX REST API and web scraped from wiki pages



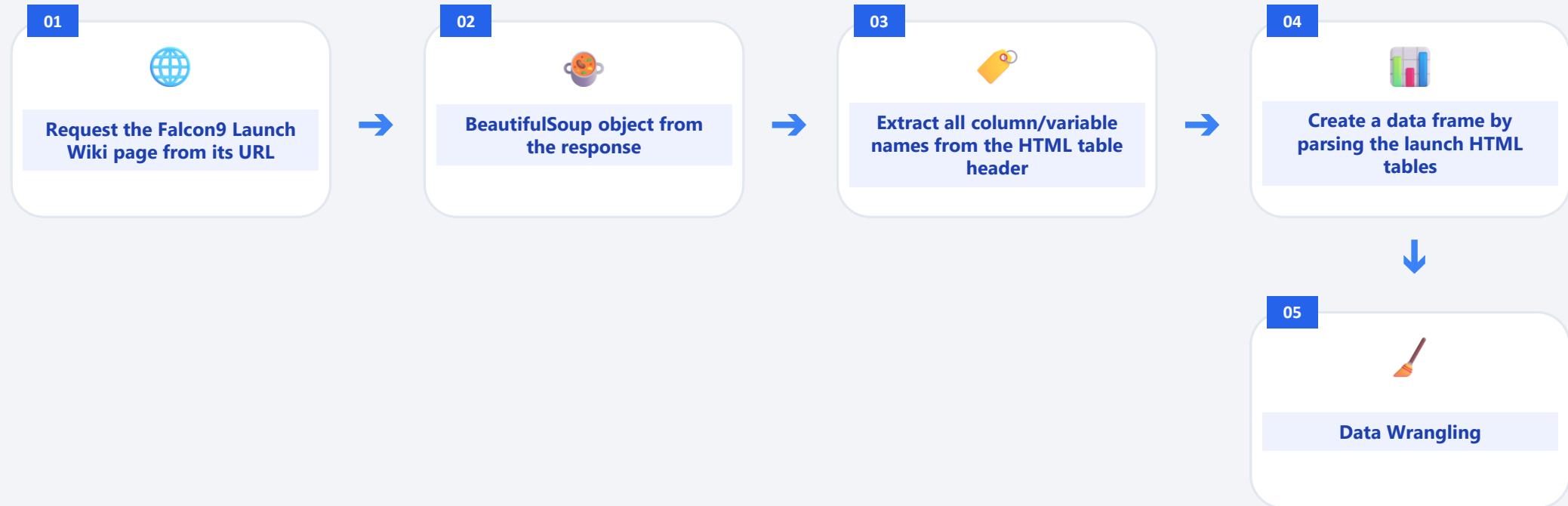
Data Collection – SpaceX Rest API

Collect and make sure the data is in the correct format from an API



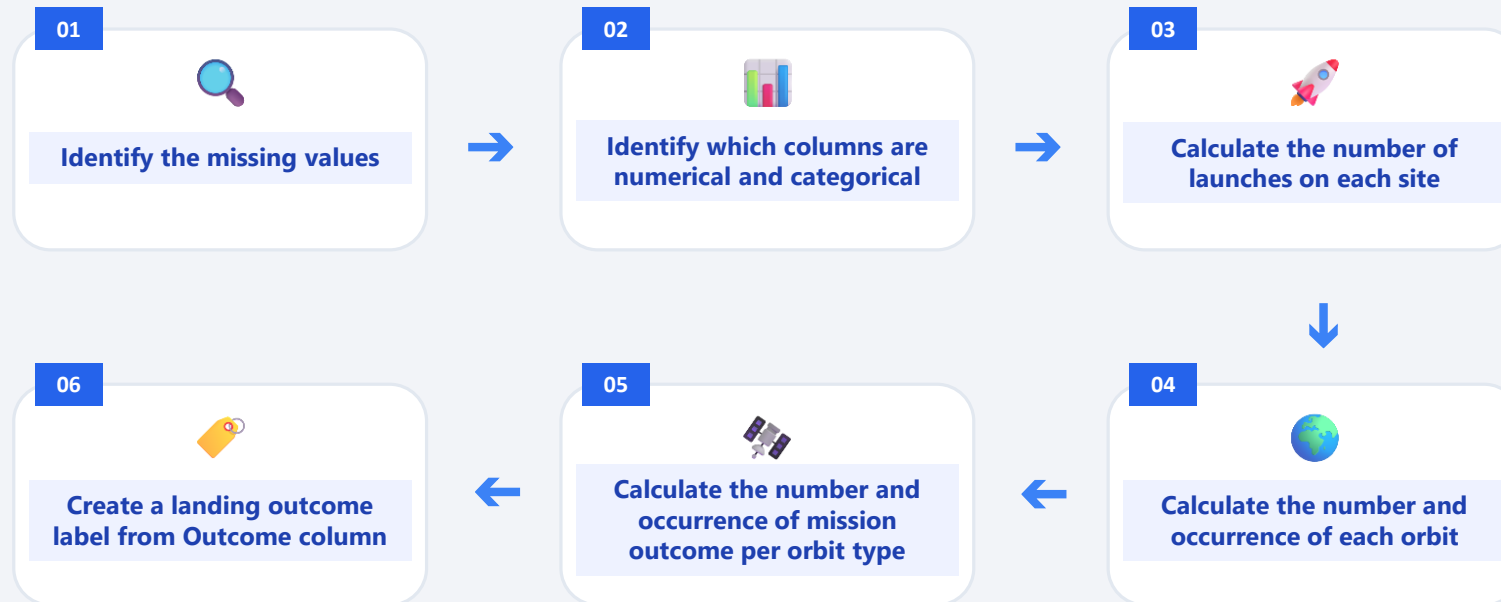
Data Collection - Scraping

Collect and make sure the data is in the correct format from and API



Data Wrangling

Perform Exploratory Data Analysis(EDA) to find patterns in the data and determine what would be the label for training the supervised models



EDA with Data Visualization

Summary of charts that were plotted

- **Catplot** to visualize the relationship between Flight Number and Payload.
- **Catplot** to visualize the relationship between Flight Number and Launch Site.
- **Catplot** to visualize the relationship between Payload and Launch Site.
- **Bar Chart** to visualize the relationship between success rate of each Orbit type.
- **Catplot** to visualize the relationship between Flight Number and Orbit type.
- **Catplot** to visualize the relationship between Payload and Orbit type.
- **Line Chart** to visualize the launch success yearly trend.

EDA with SQL

SQL queries performed:

- 1 Display the names of the unique launch sites in the space mission:

```
SELECT DISTINCT(launch_site) FROM SPACEXTBL;
```

- 2 Display 5 records where launch sites begin with the string 'CCA':

```
SELECT * FROM SPACEXTBL WHERE launch_site LIKE 'CCA%' LIMIT 5;
```

- 3 Display the total payload mass carried by boosters launched by NASA (CRS):

```
SELECT SUM(payload_mass_kg_) AS TOTAL_PAYLOAD_MASS FROM SPACEXTBL WHERE customer='NASA (CRS)';
```

- 4 Display average payload mass carried by booster version F9 v1.1:

```
SELECT AVG(payload_mass_kg_) AS AVG_PAYLOAD_MASS FROM SPACEXTBL WHERE booster_version='F9 v1.1';
```

- 5 List the date when the first successful landing outcome in ground pad was achieved:

```
SELECT MIN(DATE) AS first_successful_landing FROM SPACEXTBL WHERE (landing_outcome)='Success (ground pad)';
```

EDA with SQL

SQL queries performed:

- 6 List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000:

```
SELECT booster_version, payload_mass_kg, landing_outcome FROM SPACEXTBL WHERE landing_outcome='Success (drone ship)' AND (payload_mass_kg BETWEEN 4000 AND 6000);
```

- 7 List the total number of successful and failure mission outcomes:

```
SELECT mission_outcome, COUNT(mission_outcome) AS TOTAL FROM SPACEXTBL GROUP BY mission_outcome;
```

- 8 List the names of the booster_versions which have carried the maximum payload mass. Use a subquery:

```
SELECT DISTINCT(booster_version), (SELECT MAX(payload_mass_kg) AS "maximum_payload_mass" FROM SPACEXTBL) FROM SPACEXTBL LIMIT 5;
```

EDA with SQL

SQL queries performed:

9

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:

```
SELECT landing_outcome, booster_version, launch_site, DATE FROM SPACEXTBL WHERE landing_outcome LIKE '%Failure (drone ship)%'  
AND (DATE LIKE '2015%');
```

10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

```
SELECT landing_outcome, COUNT(landing_outcome) AS "total" FROM SPACEXTBL WHERE (DATE BETWEEN '2010-06-04' AND '2017-03-20')  
GROUP BY landing_outcome ORDER BY "total" DESC;
```

Build an Interactive Map with Folium

Summary of map objects that were created and added to the Folium map:

1 **folium.Circle** and **folium.Marker** to add a highlighted circle area with a text label on a specific coordinate for each launch site on the site map.

2 **MarkerCluster** object for simplify a map containing many markers having the same coordinate.

3 **MousePosition** on the map to get coordinate for a mouse over a point on the map.

4 **folium.PolyLine** object to draw a line between a launch site to its closest city, railway and highway.

Build a Dashboard with Plotly Dash

Summary of plots/graphs and interactions that were added to the dashboard to perform interactive visual:



Launch Site Dropdown

A launch Site Drop-down Input Component.
There are four different launch sites and a dropdown menu let us select different launch sites.



Success Pie Chart Callback

A callback function to render **success-pie-chart** based on selected site dropdown.
The general idea of this callback function is to get the selected launch site from site-dropdown and render a pie chart visualizing launch success counts.



Payload Range Slider

A range Slider to Select Payload.
The Slider is to be able to easily select different payload range and see if we can identify some visual patterns.

Success Payload Scatter Chart Callback

A callback function to render the success-payload-scatter-chart scatter plot.
To visually observe how payload may be correlated with mission outcomes for selected site(s).

Predictive Analysis (Classification)

Summary of model development process used to predict if the first stage will land given the data from the preceding labs:

1

Class → array

NumPy Array Creation

Creation of a NumPy array from the column Class in data.

2

StandardScaler()

Standardization

Data standardization.

3

80/20 split

Train / Test Split

Use of the function `train_test_split` to split the data X and Y into training and test data.

4

GridSearchCV

Hyperparameter Tuning

Searching for the best Hyperparameters for Logistic Regression, SVM, Decision Tree and KNN classifiers.

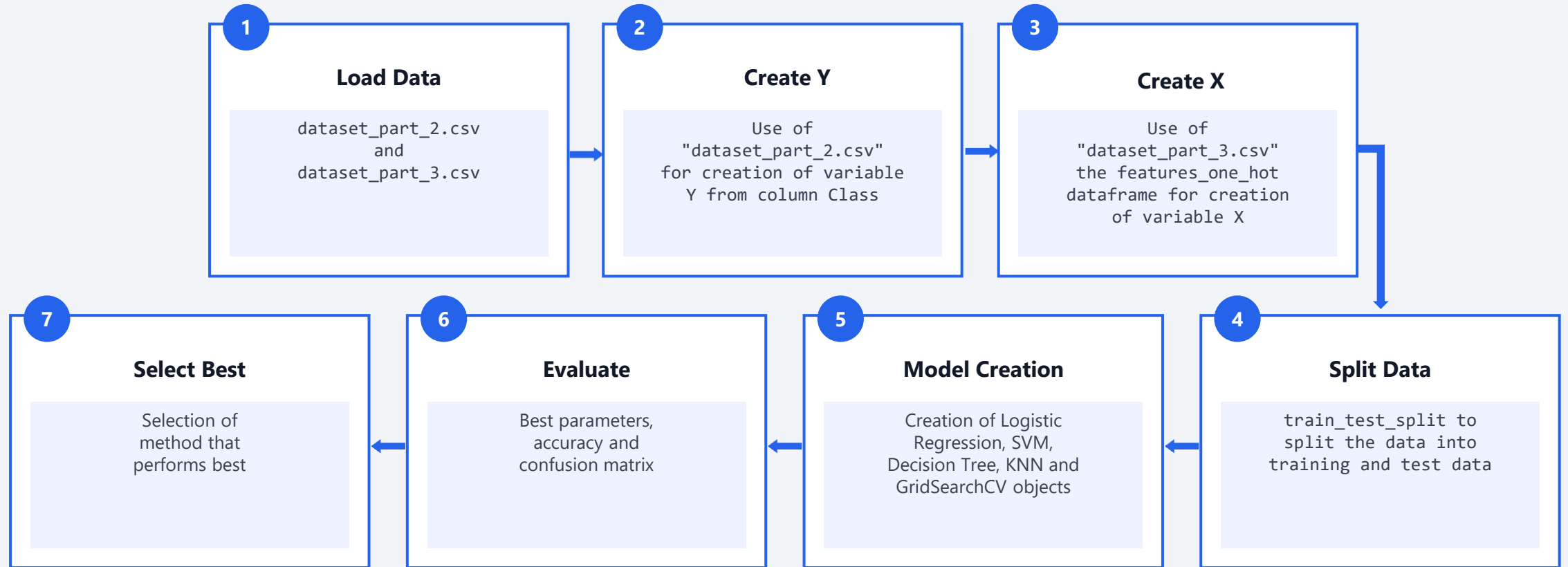
5

Compare Accuracy

Best Model Selection

Searching for the method that performs best using test data.

Predictive Analysis (Classification)

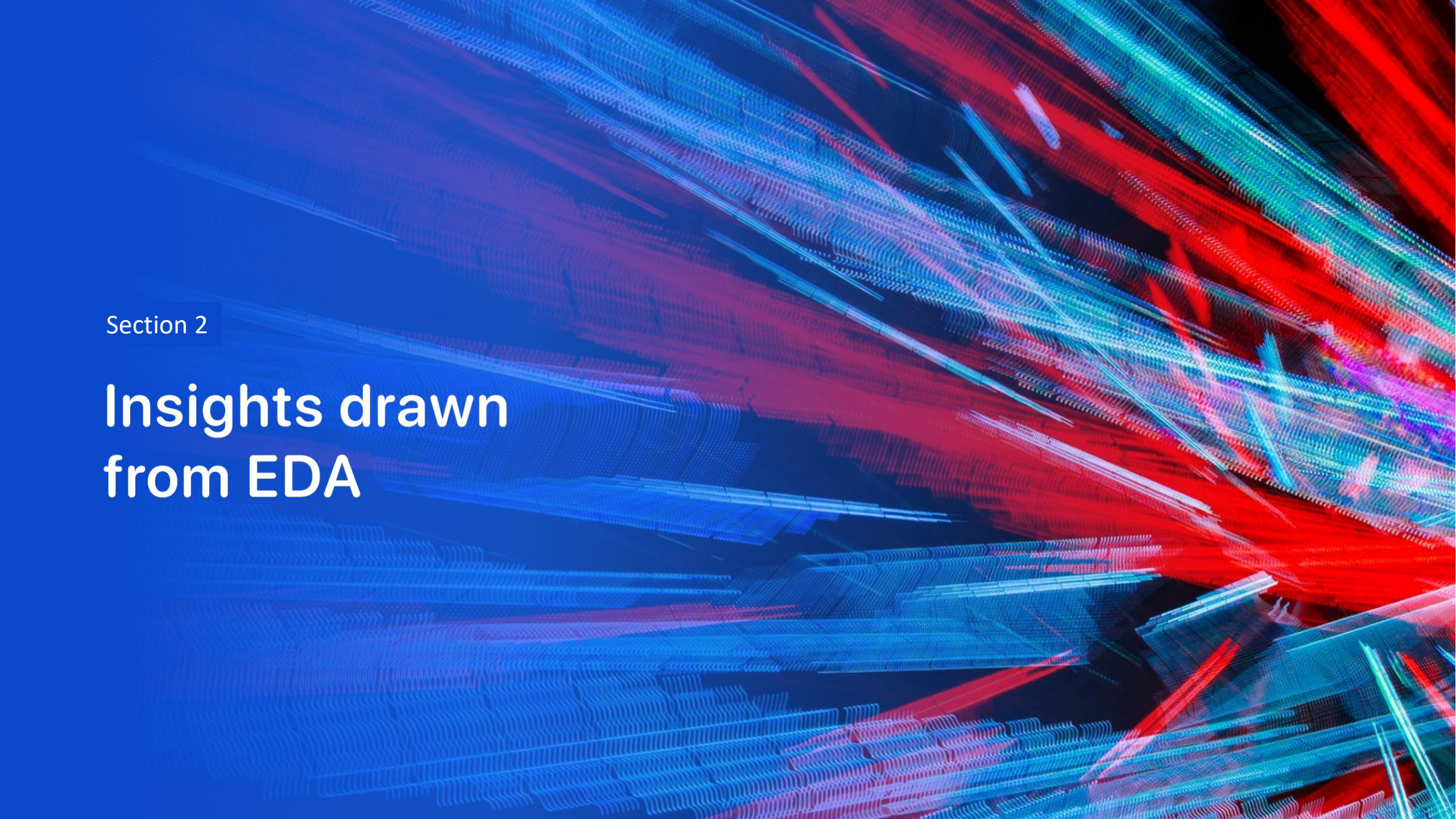


Results

1 Exploratory data analysis results

2 Interactive analytics demo in screenshots

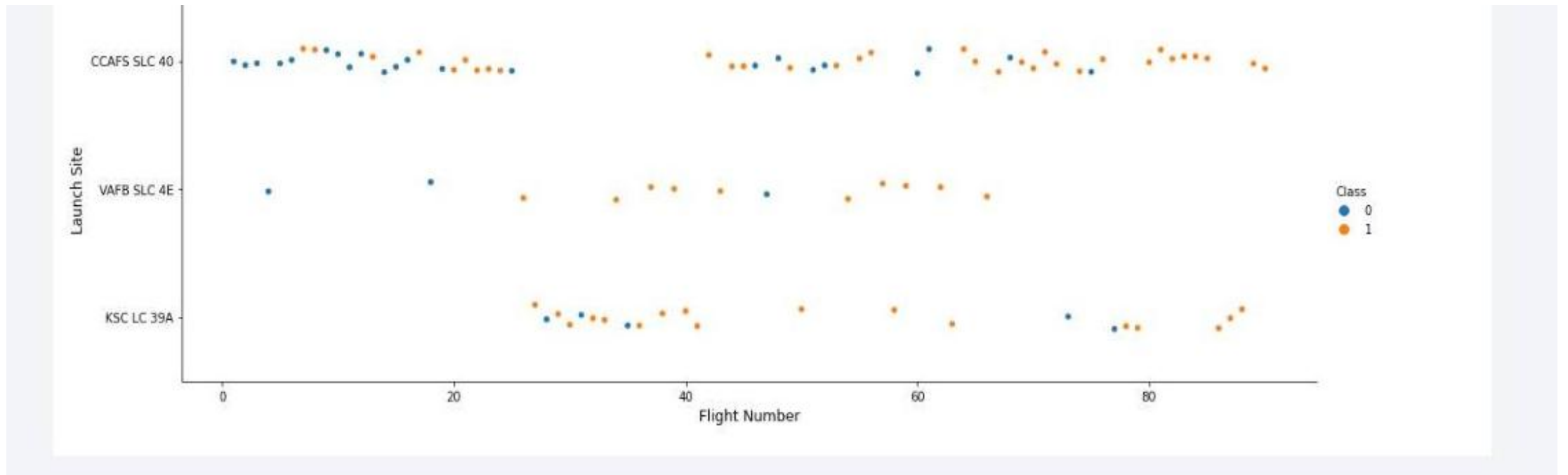
3 Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

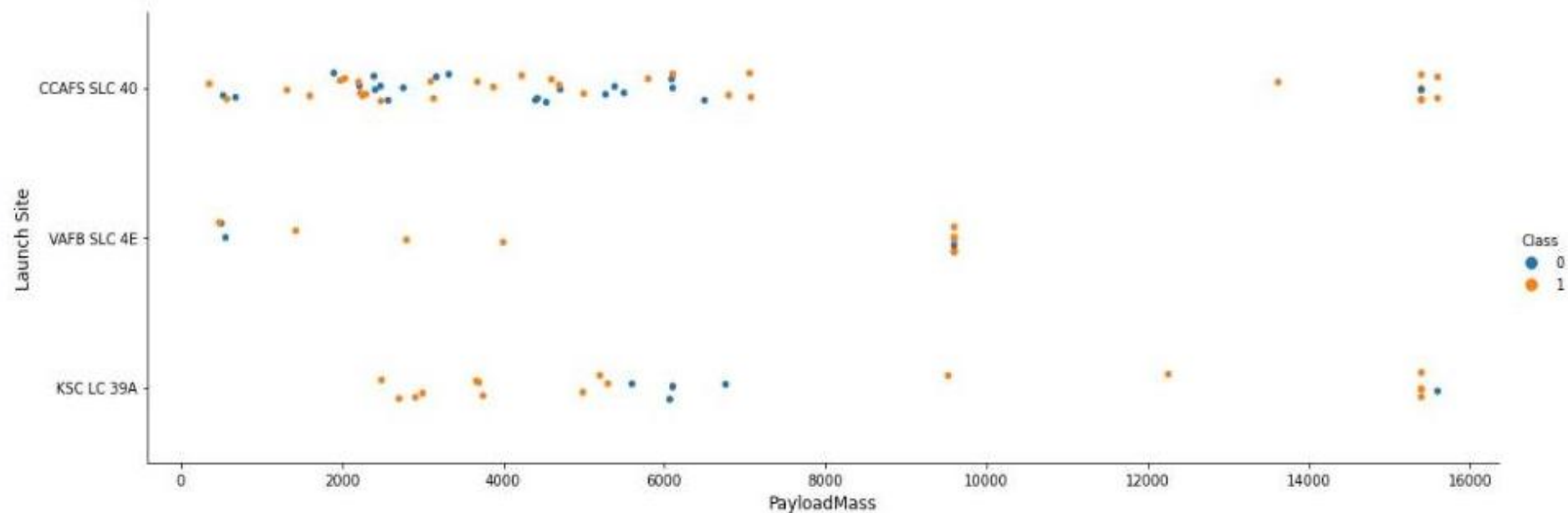
Insights drawn from EDA

Flight Number vs. Launch Site



- With time the successful rate has increased for every Launch Site, especially for CCAFS SLC 40, where are concentrated the majority of the launches.
- VAFB SLC 4E and KSC LC 39A has a higher successful rate but represents one third of the total launches.

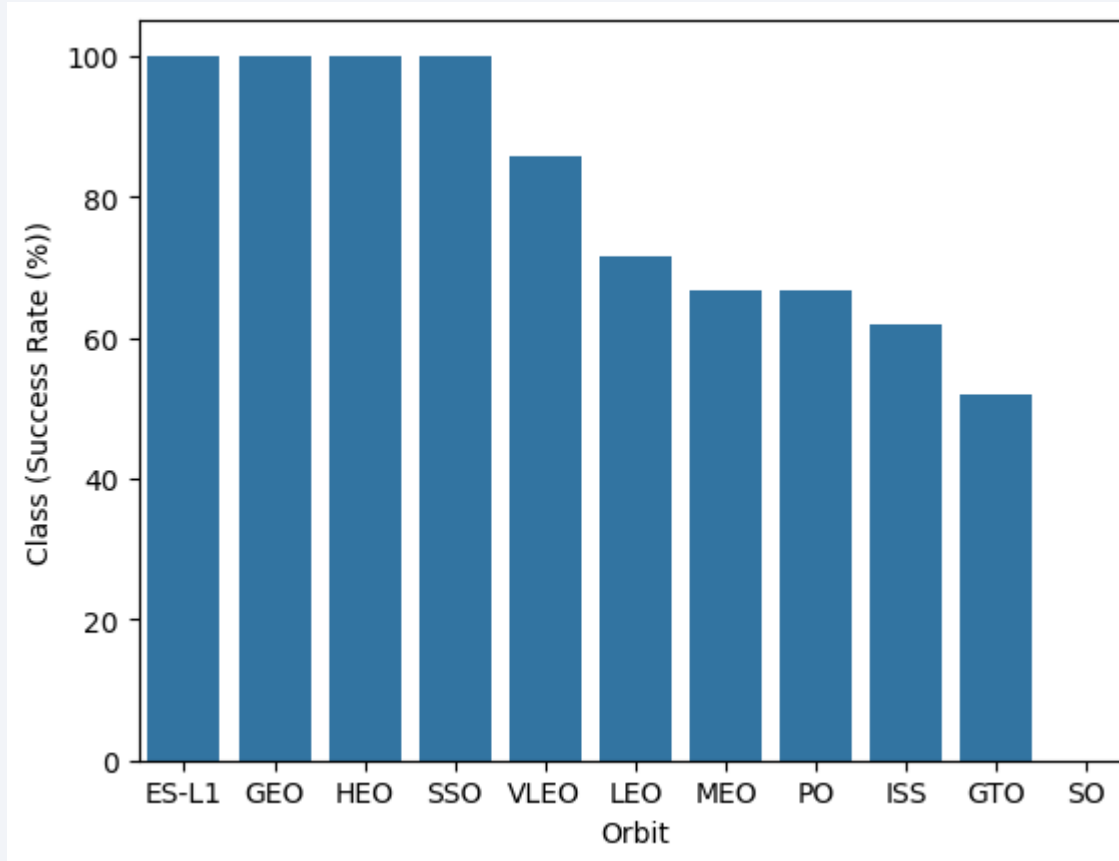
Payload vs. Launch Site



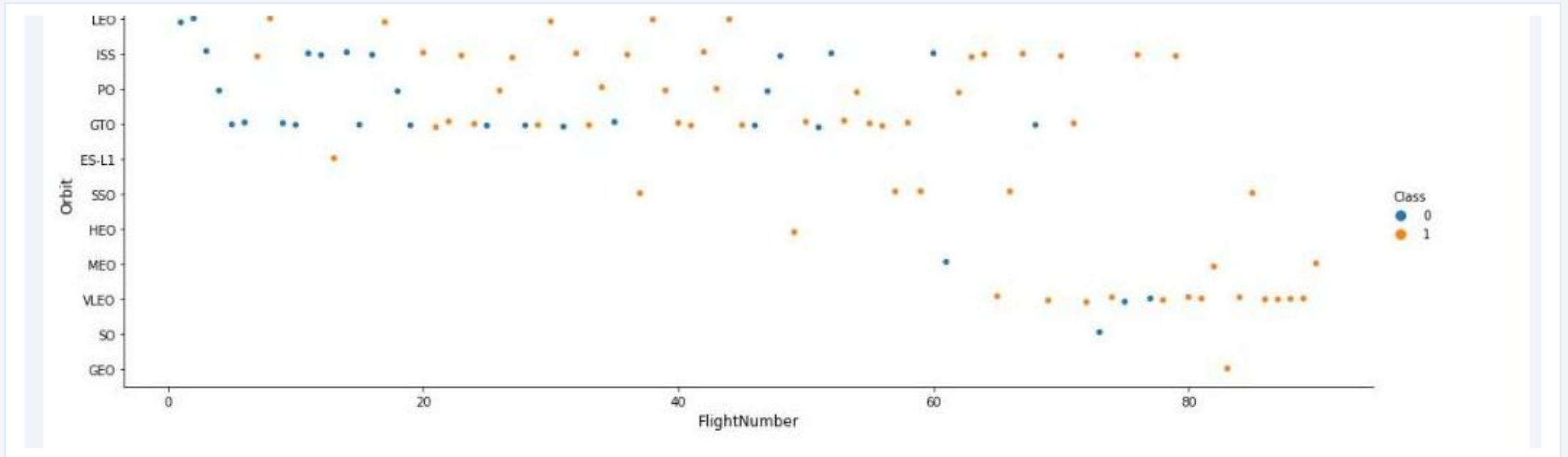
- In VAFB-SLC launch site there are no rockets launched for heavy payload mass (greater than 10000 kg).
- In KSC LC launch site there are no rockets launched for lower payload mass (less than 2500kg).
- CCAFS SLC has launched rockets less than 7500kg and more than 13000kg payload mass but not in between.

Success Rate vs. Orbit Type

- The first 4 Orbit types has the best successful rate. But how many attempts are per orbit type?
- The bar chart must be interpreted with the number of launches per orbit type.

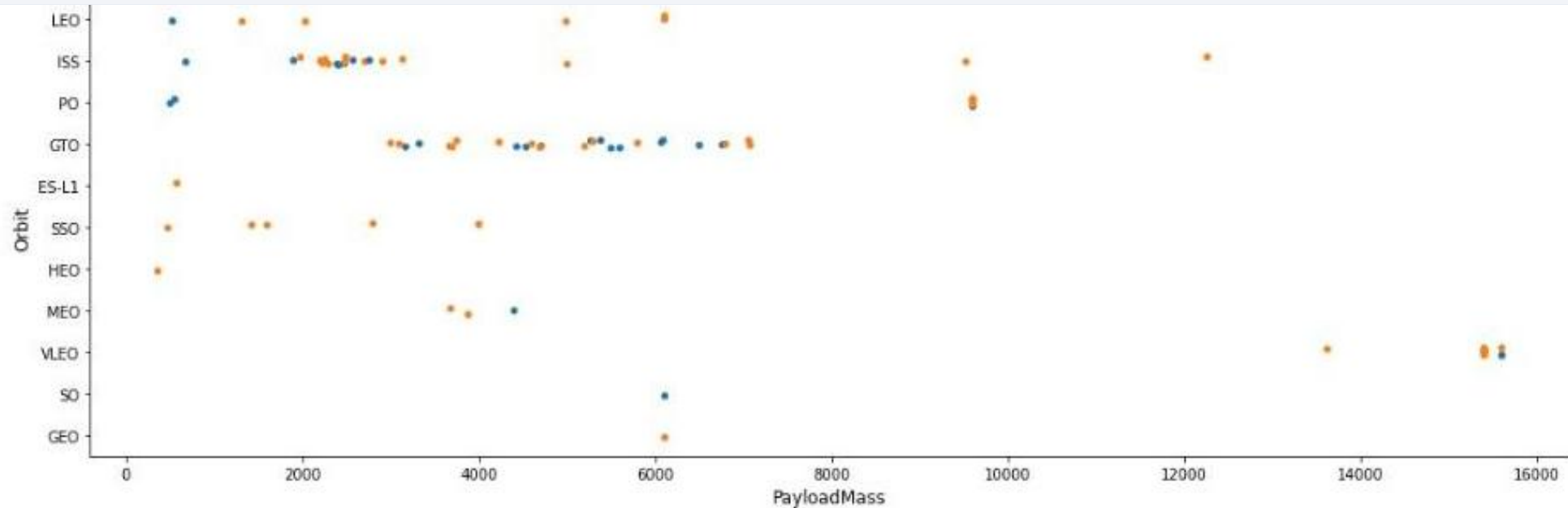


Flight Number vs. Orbit Type



- As expected, there are more failures at the beginning of the series of launches, but, after the first 40 launches, the ratio improves by reducing the 50 percent of unsuccessful landings.
- GTO and ISS orbits has the higher concentration of launches with the lowest ratio of successful landings.
- The orbits with higher successful rate, has one or just a few number of launches.

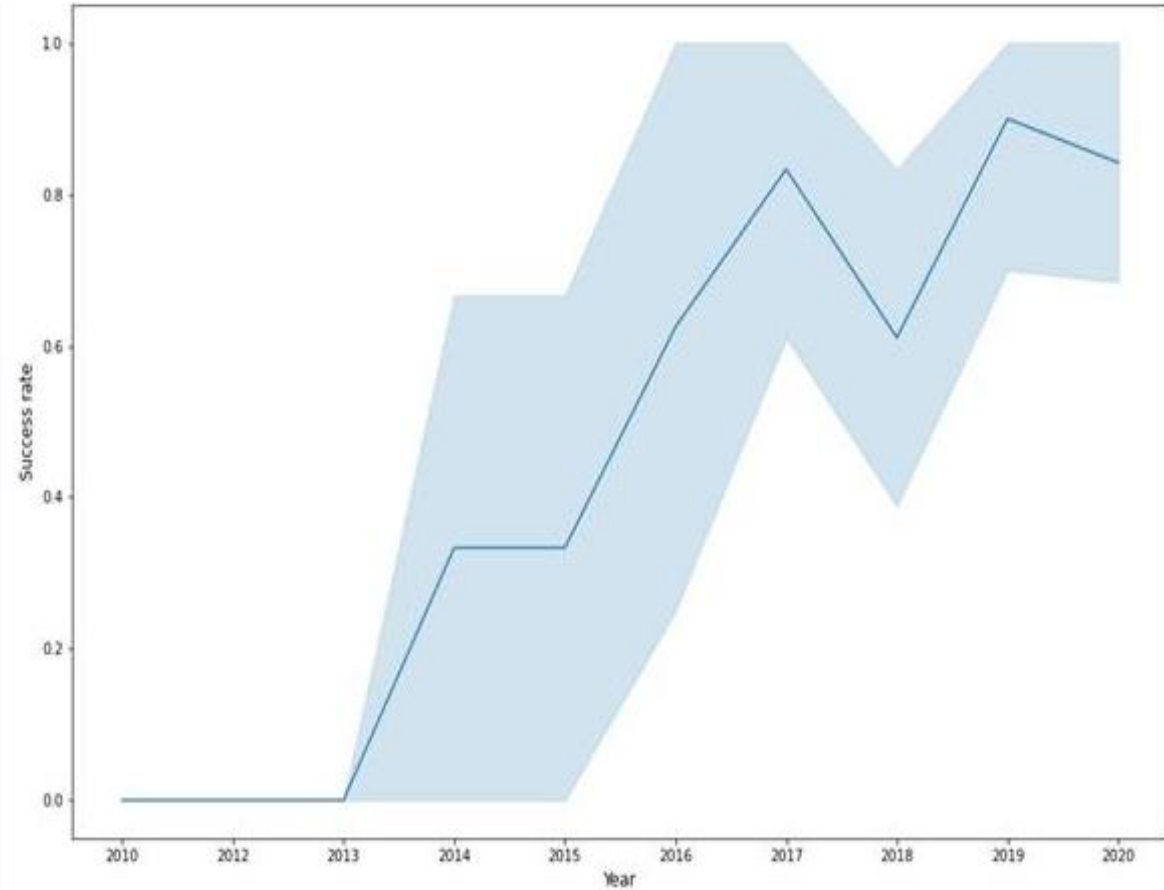
Payload vs. Orbit Type



- Exists a visible limit of Payload around 7600 kg. Less than 10 launches exceed that limit.
- With heavy payloads the successful landing rate are more for Polar, LEO and ISS.
- However for GTO, we cannot distinguish this well as both, positive landing rate and negative landing are both there here.

Launch Success Yearly Trend

- The success rate since 2013 kept increasing until 2020.



All Launch Site Names

- The four unique launch sites in the space mission.
- I have used "DISTINCT" statement to find the unique values in the launch site column.

```
%sql SELECT DISTINCT(launch_site) as "Launch_Sites" FROM SPACEXTBL;
```

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

Launch Site Names Begin with 'CCA'

5 records where launch sites begin with the string 'CCA'. The query uses WHERE, LIKE and LIMIT.

```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

The total payload mass carried by boosters launched by NASA (CRS) using SUM function and WHERE clause.

```
%SELECT SUM(PAYLOAD_MASS_KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

Total_Payload_Mass(kgs)	Customer
45596	NASA (CRS)

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 using AVG() function.

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1%';
```

Payload Mass Kgs	Customer	Booster_Version
2534.6666666666665	MDA	F9 v1.1 B1003

First Successful Ground Landing Date

The date when the first successful landing outcome in ground pad was achieved using MIN function.

```
%sql SELECT MIN(DATE) FROM SPACEXTBL WHERE 'Landing_outcome' = 'Success (ground pad)';
```

MIN(DATE)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000, combining WHERE clause with AND operator.

```
%sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;
```

booster_version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11/EchoStar 105

Total Number of Successful and Failure Mission Outcomes

The total number of successful and failure mission outcomes. The query uses a combination of COUNT function with GROUP BY statement.

```
%sql SELECT 'Mission_Outcome', COUNT('Mission_Outcome') AS TOTAL FROM SPACEXTBL GROUP BY 'Mission_Outcome';
```

Mission_Outcome	Total
Failure (in flight)	1
Success	99
Success (Payload status unclear)	1

Boosters Carried Maximum Payload

The names of the booster versions which have carried the maximum payload mass. Using a subquery.

```
%sql SELECT "Booster_Version",Payload, "PAYLOAD_MASS_KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTBL);
```

Booster_Version	Payload	PAYLOAD_MASS_KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

2015 Launch Records

The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql SELECT lansql SELECT substr(DATE, 6, 2) AS Month_Number, substr(DATE, 1, 4) AS Year, "Booster_Version", "Launch_Site", "Payload", "PAYLOAD_MASS_KG_", "Mission_Outcome",  
"Landing_Outcome" FROM SPACEXTBL WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr(DATE, 1, 4) = '2015';
```

Month_Number	Year	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Mission_Outcome	Landing_Outcome
01	2015	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	Success	Failure (drone ship)
04	2015	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	Success	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order. The query uses COUNT, WHERE, BETWEEN and GROUP BY.

```
%sql SELECT * FROM SPACEXTBL WHERE "Landing_Outcome" LIKE 'Success%' AND (Date BETWEEN '2010-06-04' AND '2017-03-20') ORDER BY Date DESC;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-01-14	17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communications	Success	Success (drone ship)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-07-18	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2016-05-27	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)
2016-05-06	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-04-08	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

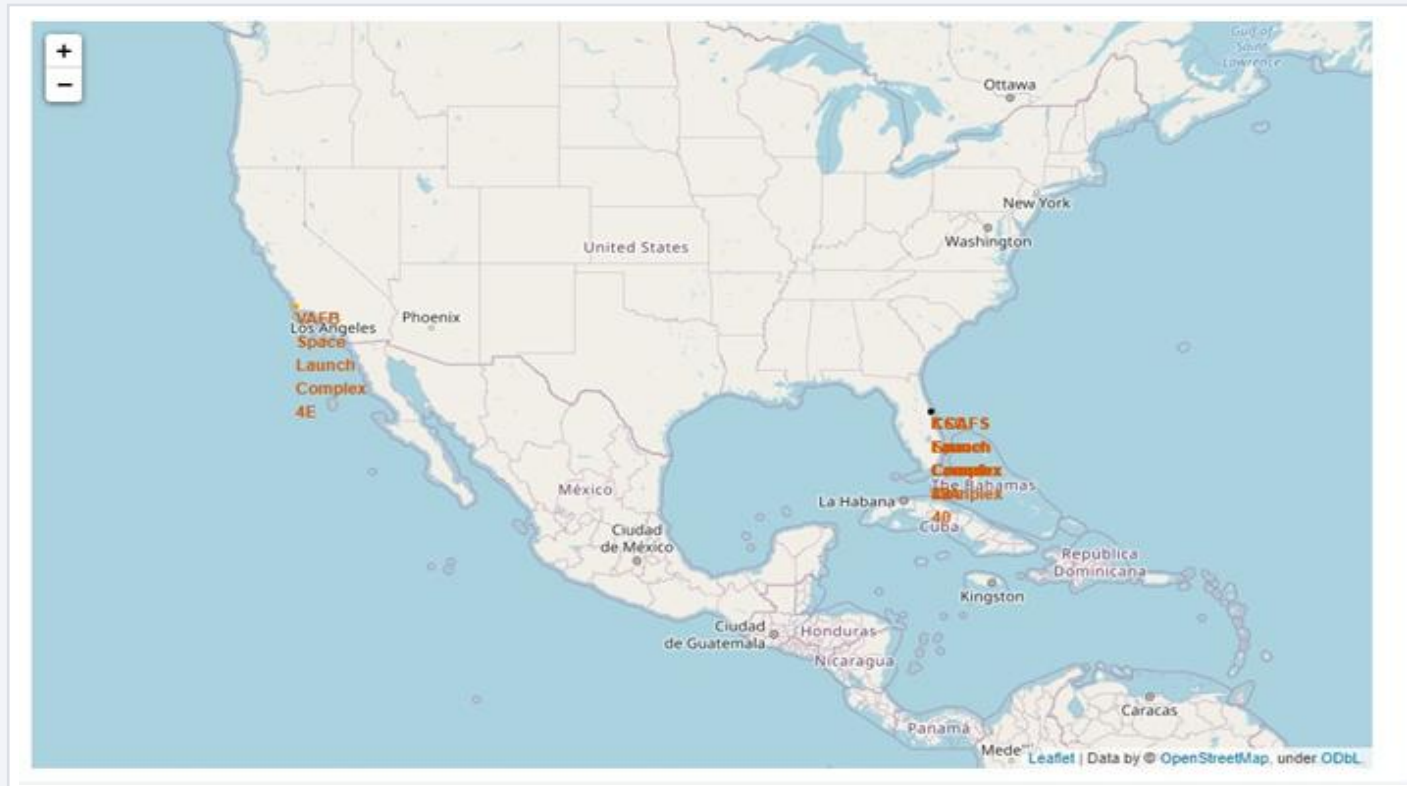
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

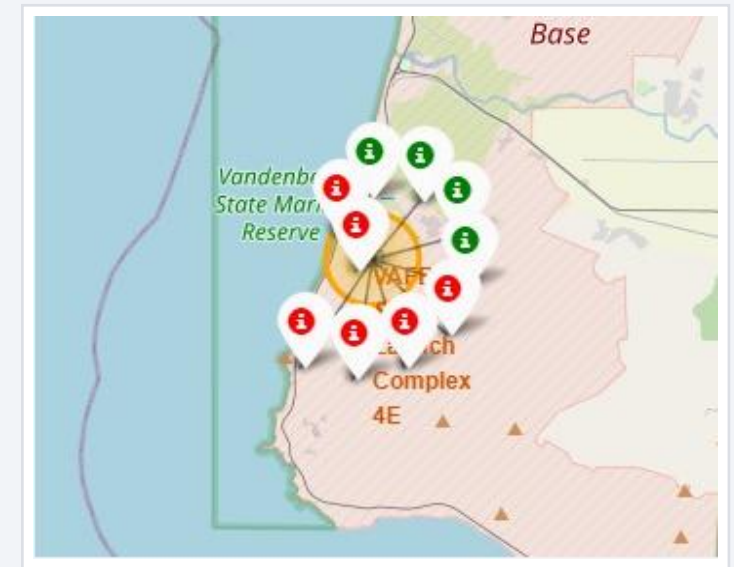
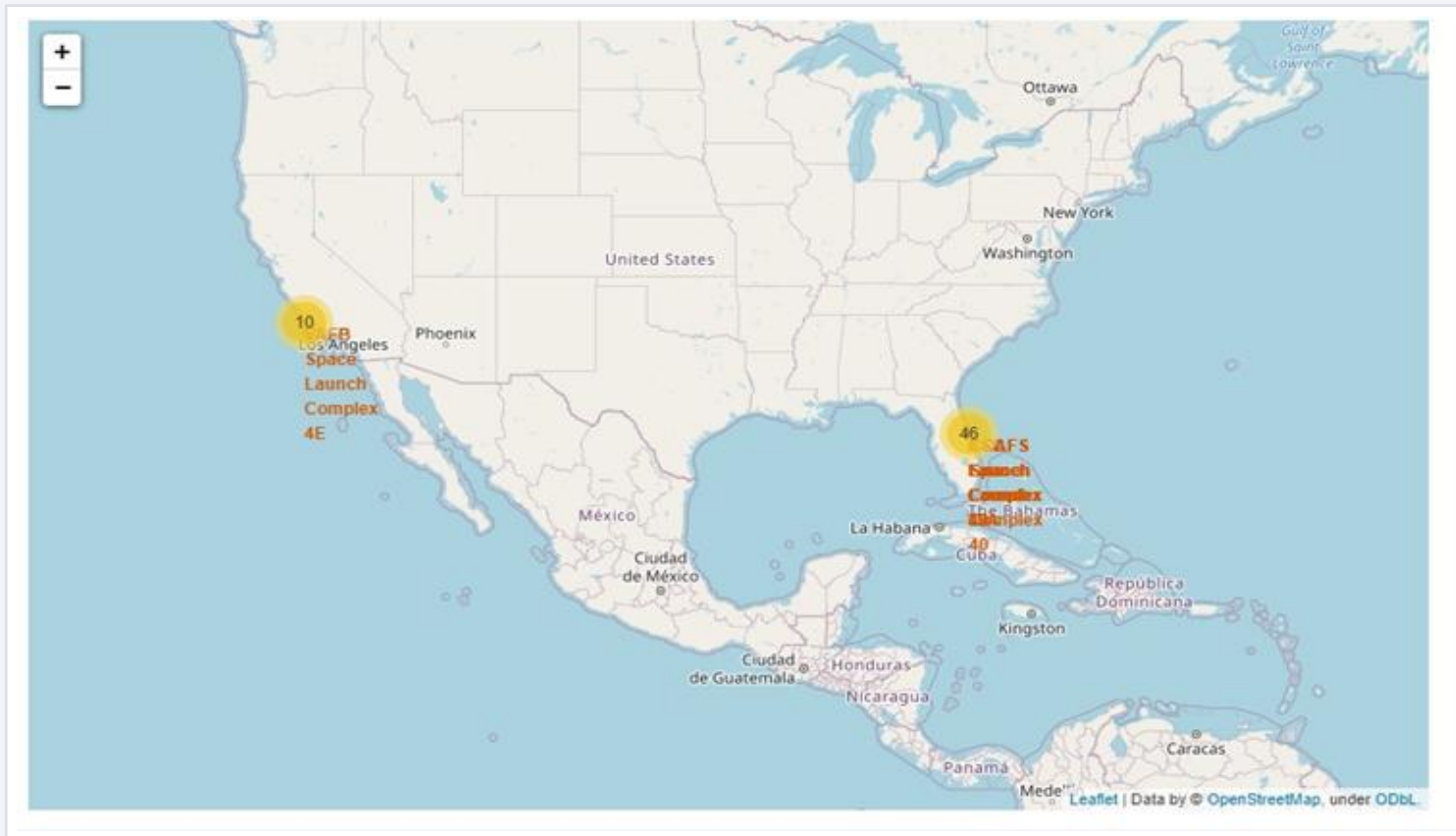
<All Launch Sites>

All launch sites are in very close proximity to the coast and into restricted areas.



<Success/Failed Launches For Each Site>

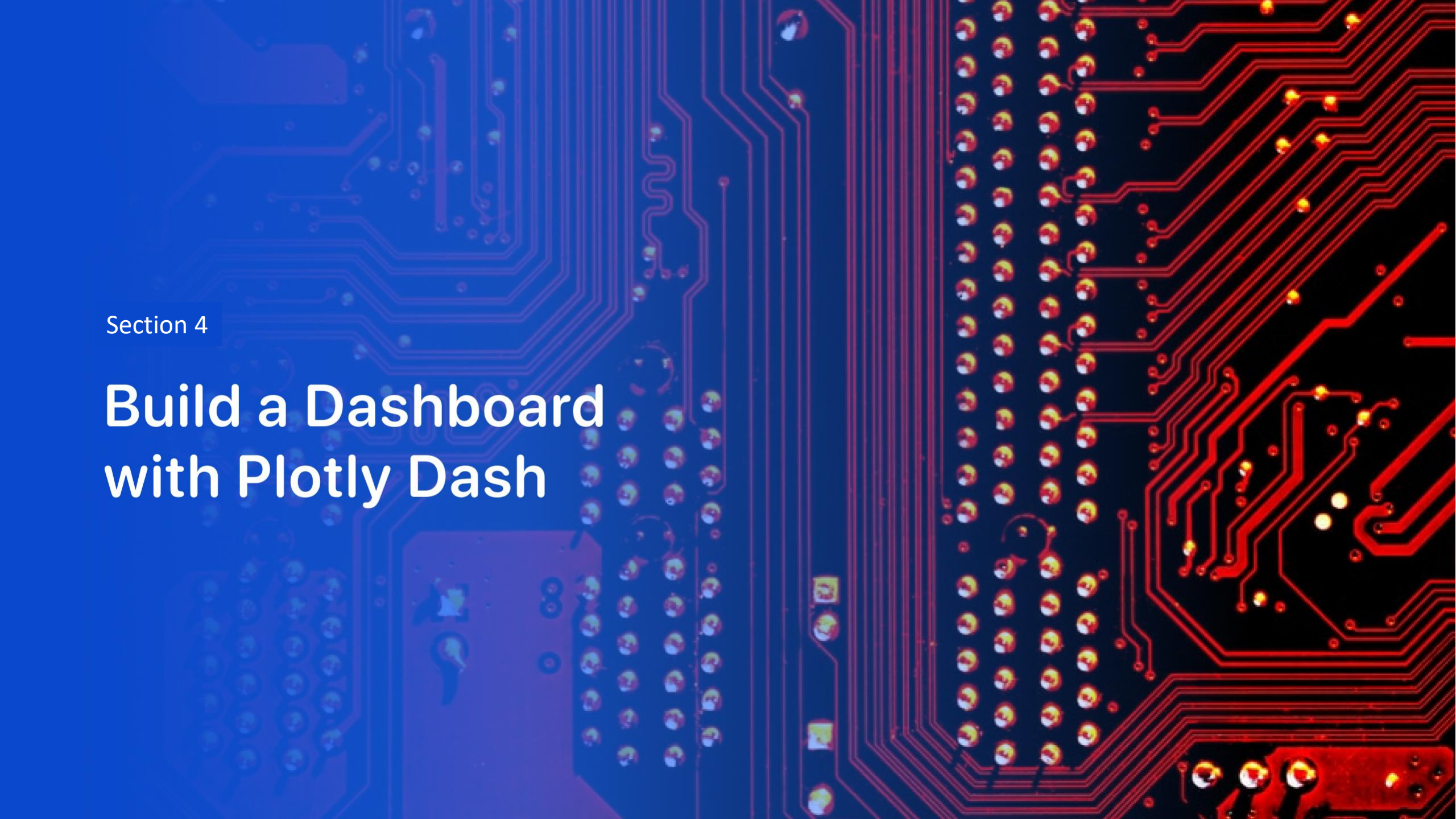
The first map shows clusters for every launch site, the second shows a green marker if a launch was successful, and a red marker if a launch was failed.



<Launch Site And Its Proximities>

Launch sites are near to railways, roads, highways and coastline. I understand that it is not just for easy supply or access but, for maintain a safe distance with near cities.



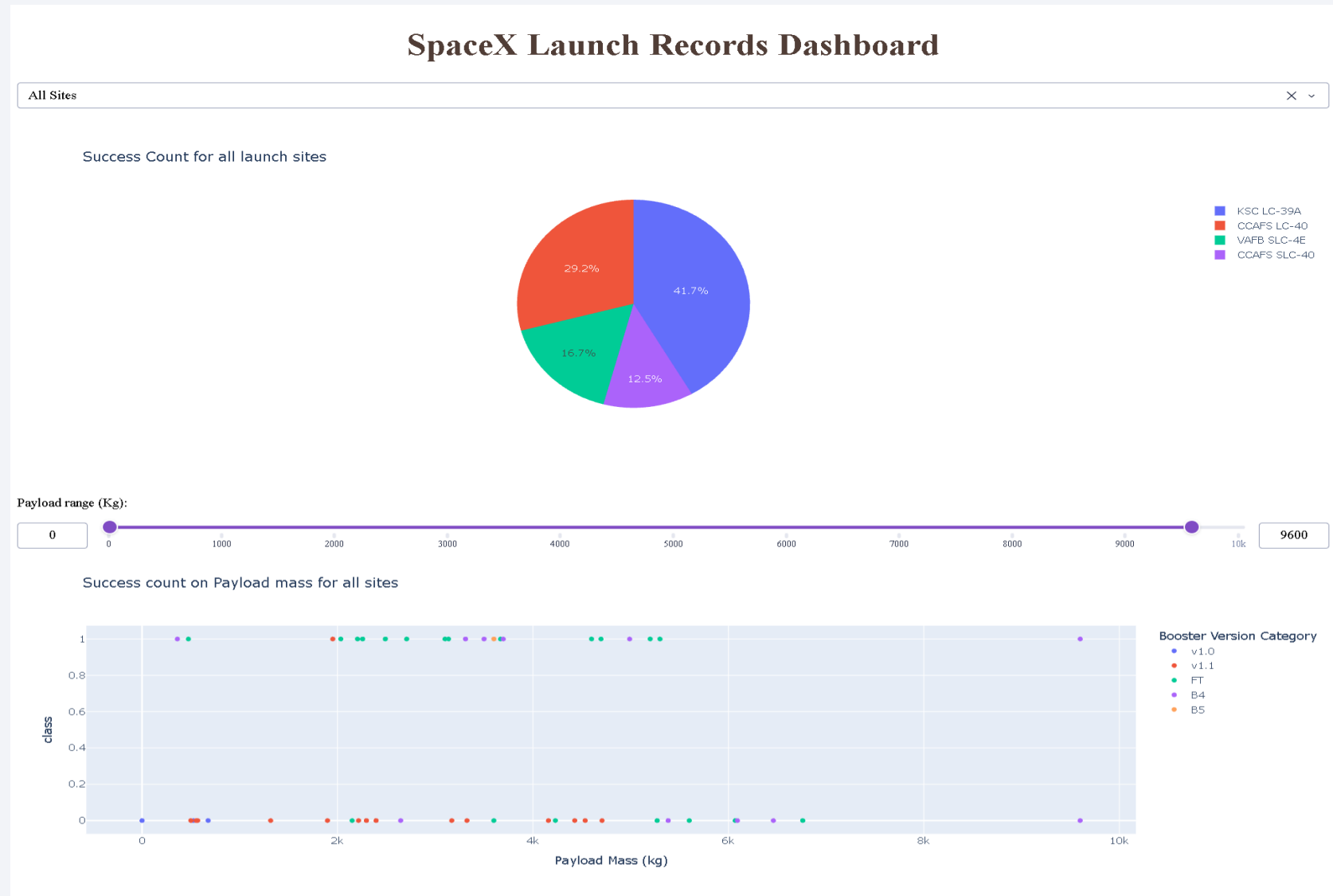


Section 4

Build a Dashboard with Plotly Dash

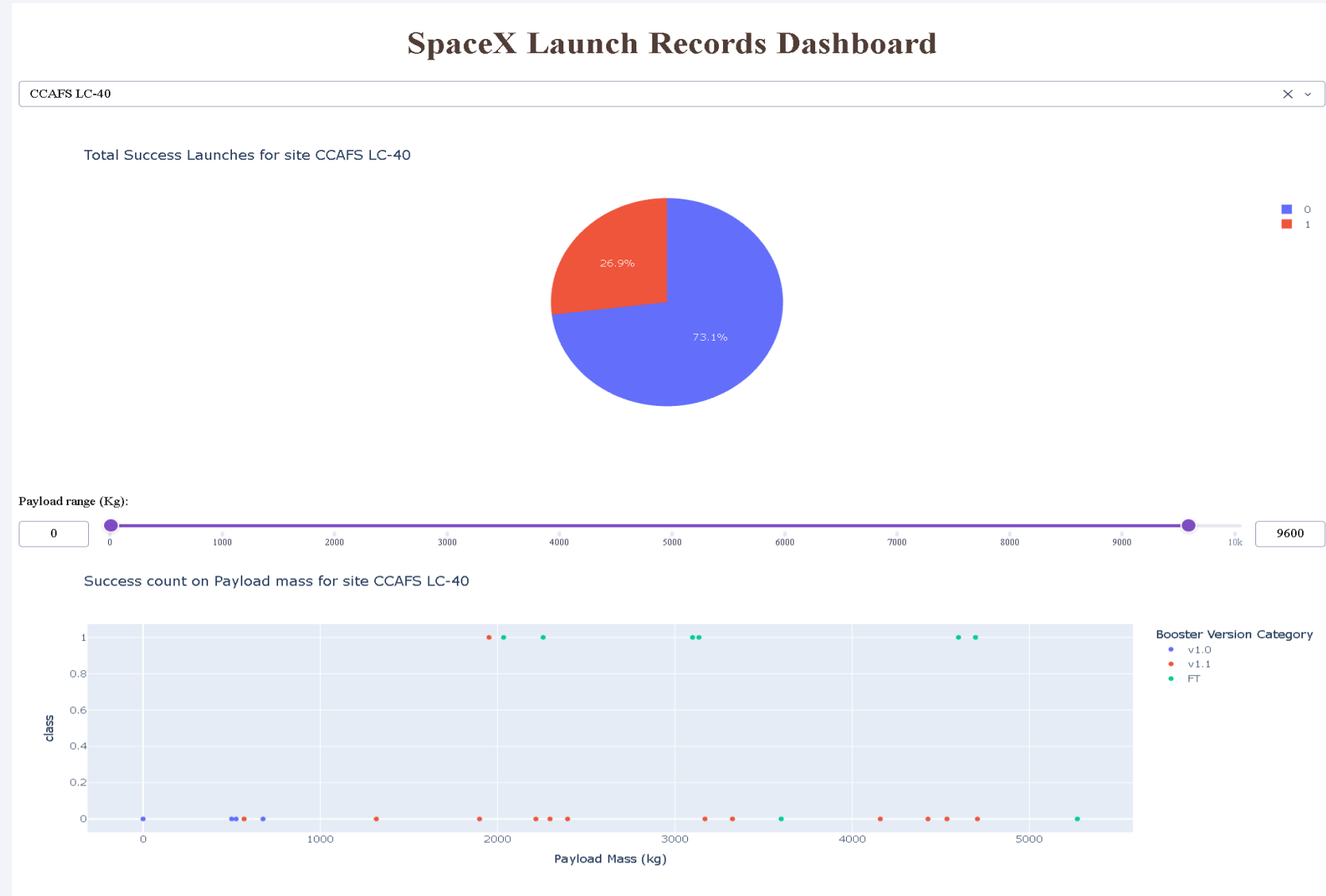
<Dashboard Screenshot 1>

- Pie chart for success count of all the launch sites.
- Scatter plot for success count on payload mass for all the sites of all the Booster version category



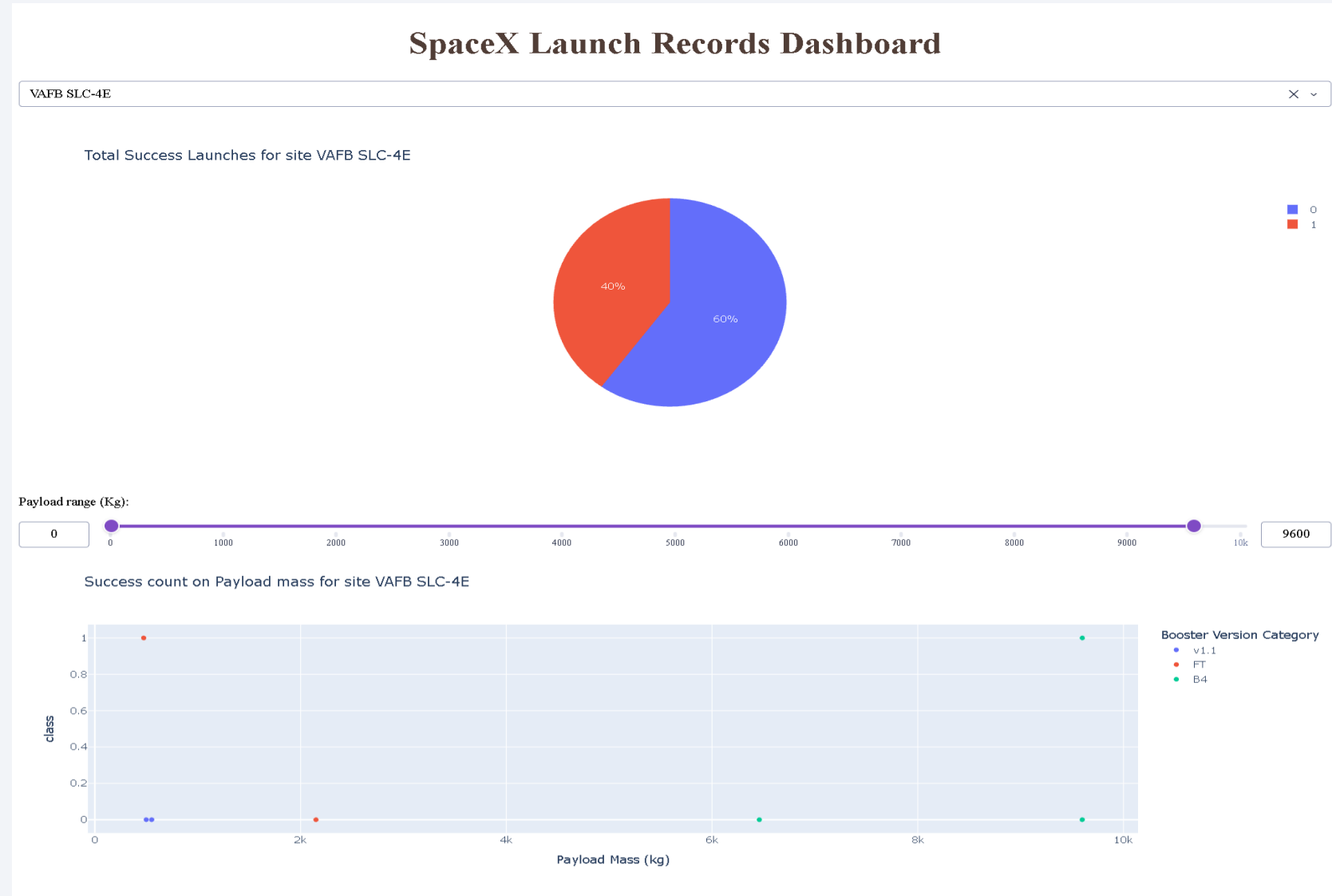
<Dashboard Screenshot 2>

- Pie chart for success count of the launch site CCAFS LC-40
- Scatter plot for success count on payload mass for the site CCAFS LC-40 of different Booster version category



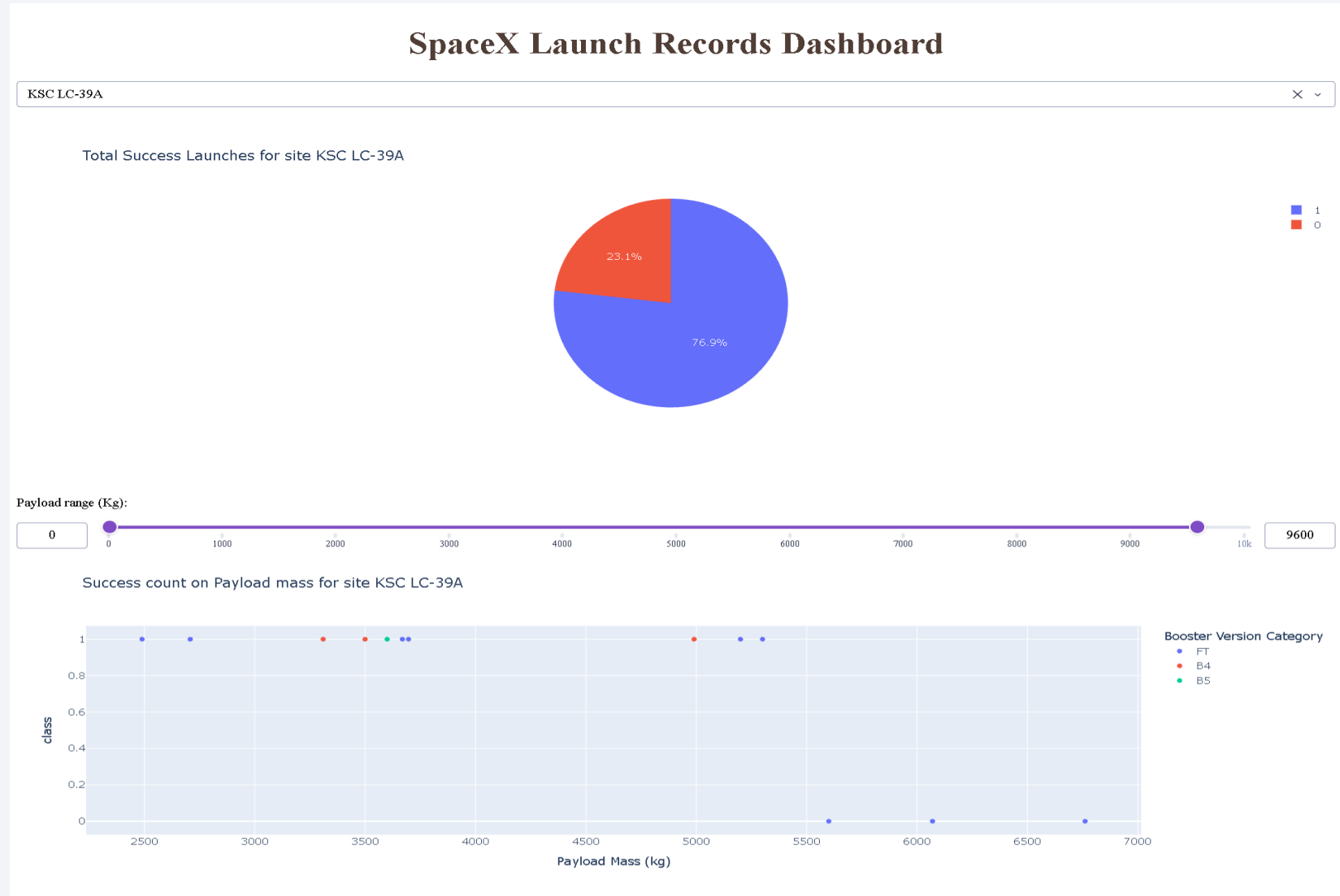
<Dashboard Screenshot 3>

- Pie chart for success count of the launch site VAFB SLC-4E
- Scatter plot for success count on payload mass for the site VAFB SLC-4E of different Booster version category



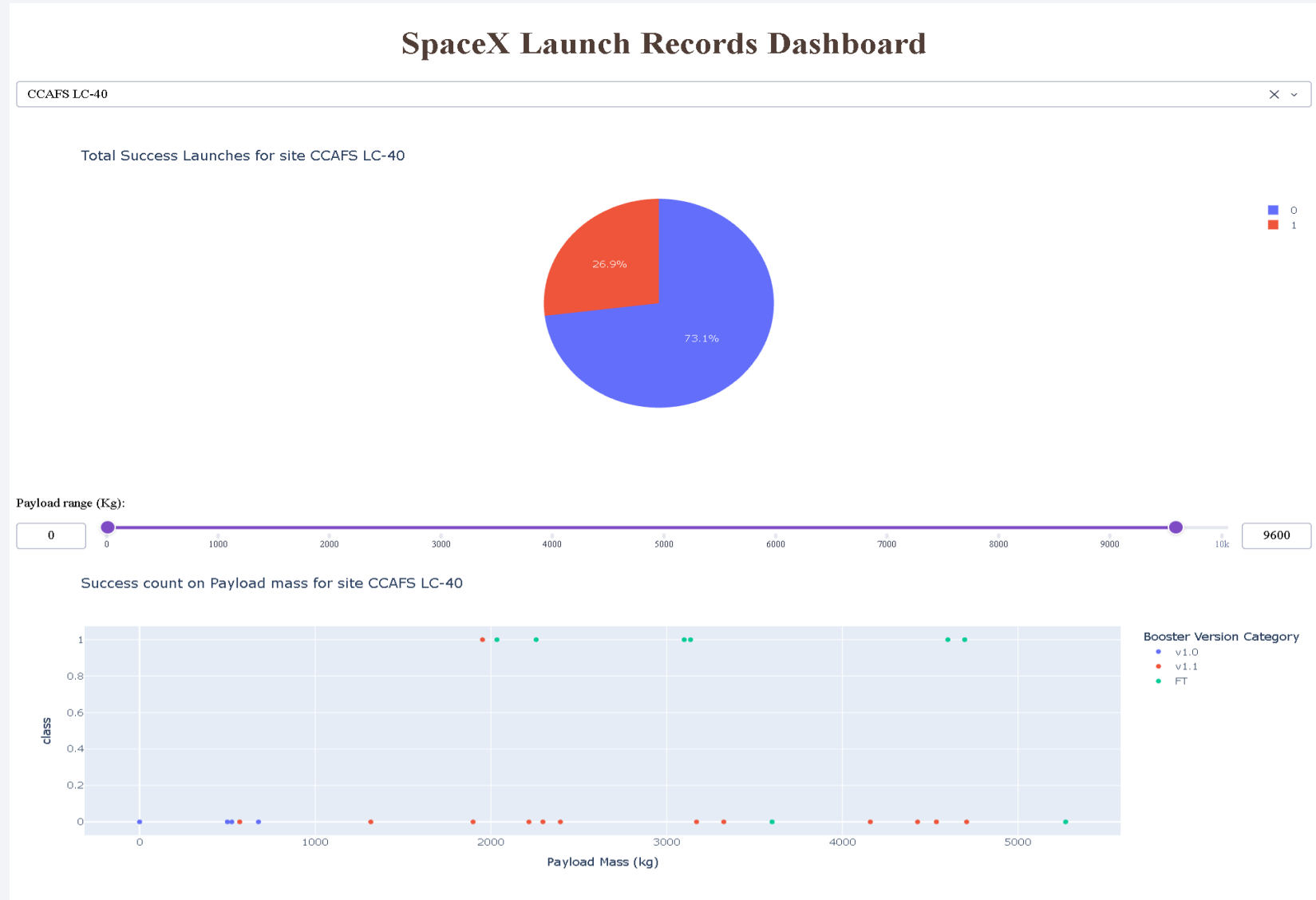
<Dashboard Screenshot 4>

- Pie chart for success count of the launch site KSC SLC- 39A
- Scatter plot for success count on payload mass for the site KSC SLC- 39A of different Booster version category



<Dashboard Screenshot 5>

- Pie chart for success count of the launch site CCAFS LC-40
- Scatter plot for success count on payload mass for the site CCAFS LC-40 of different Booster version category



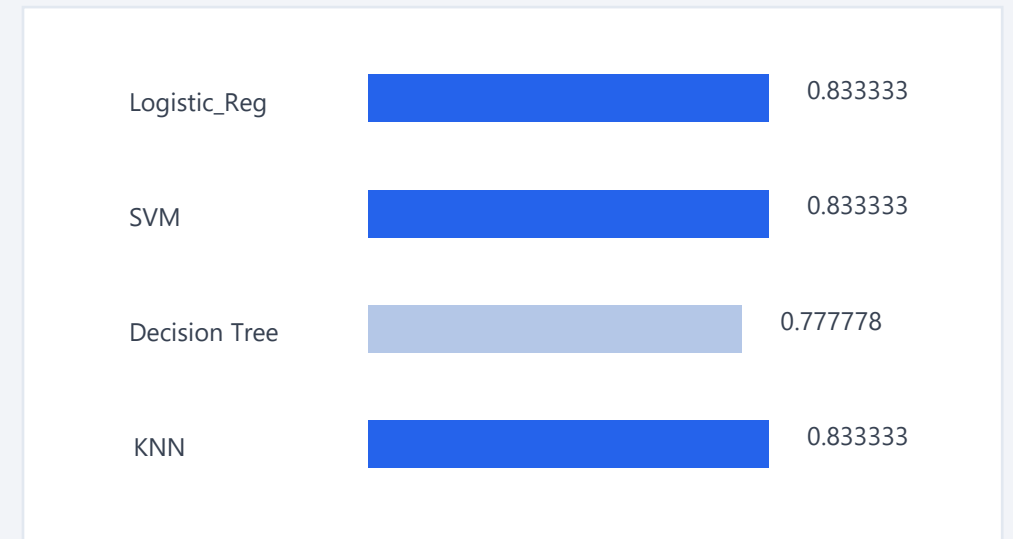
Section 5

Predictive Analysis (Classification)

Classification Accuracy

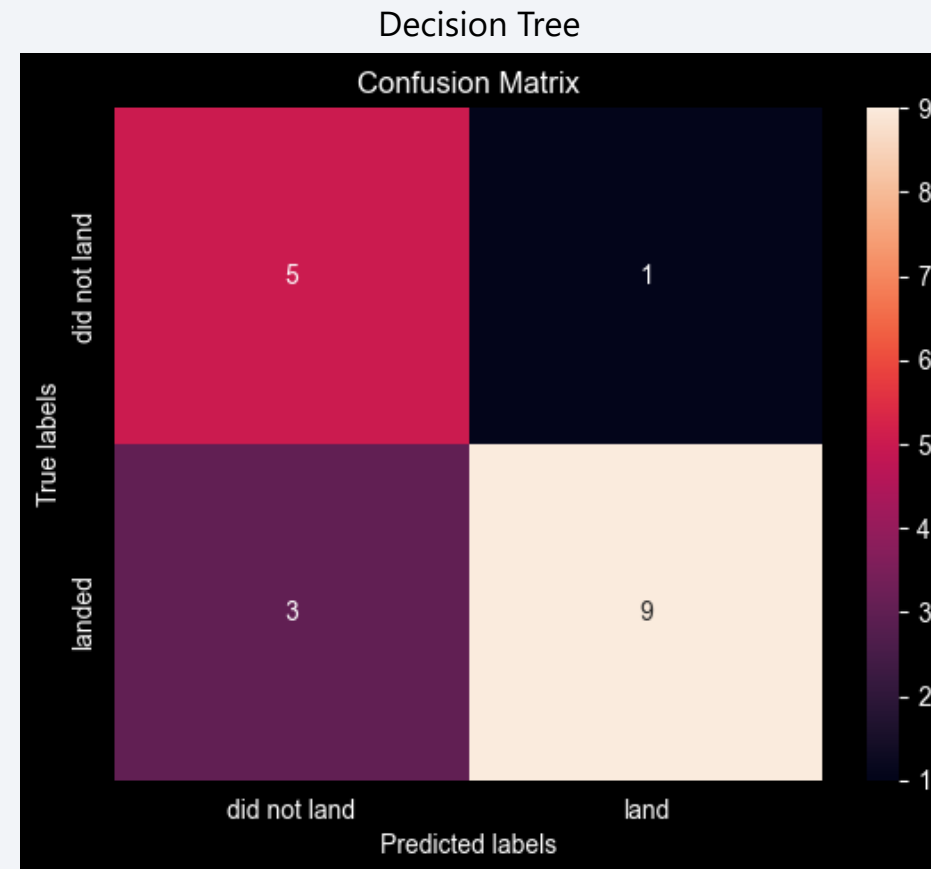
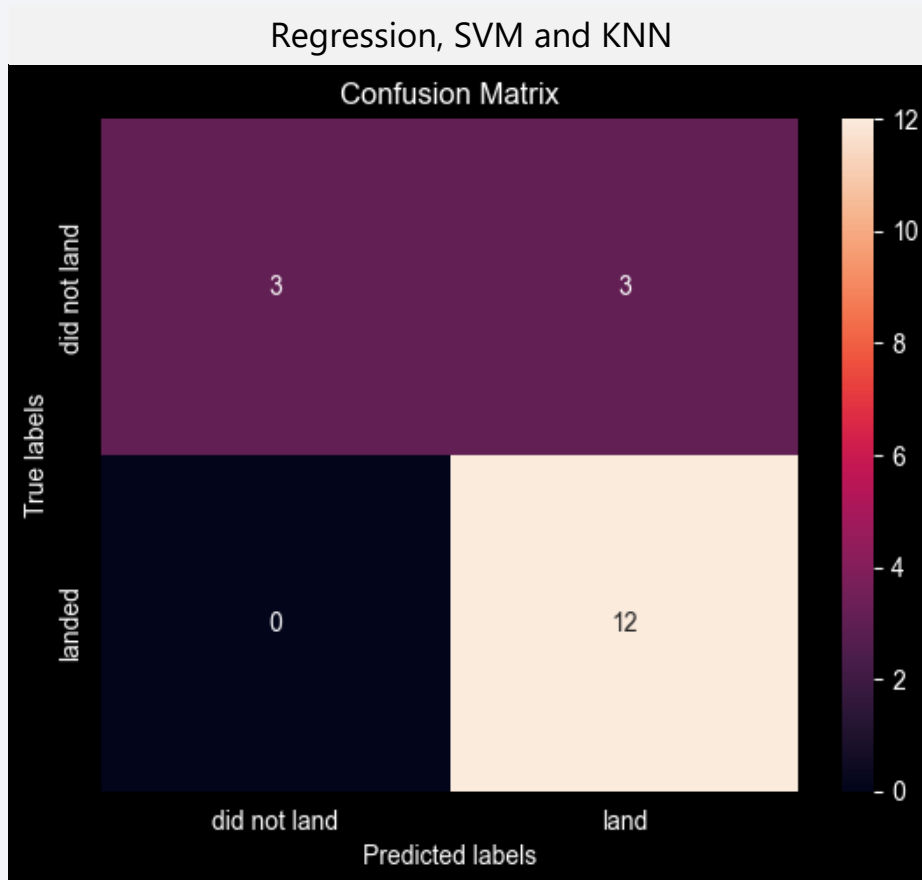
The accuracy is same for all the models (Regression, SVM and KNN) is 0.83 whereas only Decision Tree model has the accuracy of 0.78

Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.777778
KNN	0.833333



Confusion Matrix

Confusion matrix is same for all the models (Regression, SVM and KNN) and only Decision Tree model has different Confusion matrix



Conclusions

- 1 More launches at a site mean higher success rates.
- 2 Success rates rose from 2013 to 2020.
- 3 Orbits ES-L1, GEO, HEO, SSO, and VLEO had top success.
- 4 KSC LC-39A led in successful launches.
- 5 Decision tree classifier is the best ML model for this task.

Appendix

Follow the Github repository link for Dataset and Notebooks as below:

Applied Datascience Capstone

Thank you!

